

RESEARCH ARTICLE

Design of an Optimized Feature Driven Severity Stage Classifier for Parkinson's Disease Prediction Using Deep Learning

K. SHYAMALA^{ID} AND T. M. NAVAMANI^{ID}

School of Computer Science and Engineering (SCOPE), Vellore Institute of Technology (VIT), Vellore, Tamil Nadu 632 014, India

Corresponding author: T. M. Navamani (navamani.tm@vit.ac.in)

ABSTRACT Parkinson's Disease (PD) is a degenerative neurological condition that seriously affects both motor and non-motor abilities. Among many biomarkers, speech abnormalities have become a potential sign of PD as the disease affects the motor control of the vocal point. Therefore, speech data is a convenient, non-invasive, and valuable diagnostic tool. This work presents a novel methodology for classifying the severity stage of parkinson's disease using speech data, addressing a notable deficiency in existing research works that have focused on binary classification. Stage classification distinguishes PD's early, mid, and late phases, offering essential insights for stage-specific therapy strategies. We proposed a Parkinson's Speech Stage-Net model (PSS-Net) for the prediction of the severity of PD that includes Isolation Forest-based outlier detection and removal, KMeansSMOTE data balancing, and Optimized Dimensionality Reduction techniques. The essential speech features were found using a tree-based feature selection method. Selected features are fed into the Optimized and Fused Neural Network model. The proposed model effectively classified the stages of PD with an accuracy of 89%. Moreover, this research study stresses openness using SHAP (SHapley Additive exPlanations) to understand feature contributions, thereby improving the clinical relevance and credibility of the model.

INDEX TERMS Parkinson's disease, deep learning, explainable artificial intelligence, stage classification, speech biomarker, KMeansSMOTE.

I. INTRODUCTION

Parkinson's Disease (PD) is a neurodegenerative condition severely impacting human neurological function [1], [2], [3]. According to current survey results [4], [5], PD is the second most frequent disease among many individuals worldwide and results from the nervous system's continuous loss of dopamine neurons. Furthermore, this condition has to be detected in the early phases. Therefore, one of the challenges is that a suitable therapy for it has yet to be provided [6], [7], [8]. Analysis of speech sounds is a practical, non-invasive approach for PD detection. Non-invasive PD detection and prediction techniques attract doctors and neuroscientists. Furthermore, identifying speech changes in the affected

roles [9] might help in early identification and interference before the beginning of physically debilitating signs, thus affecting the patient's quality of life and the medical system. Table 1 shows the PD stages with a description.

The Unified Parkinson's Disease Rating Scale (UPDRS) is a tool used to evaluate and monitor the severity and progression of PD. It assesses motor and non-motor symptoms to deliver a holistic view of the patient's condition. The scale helps clinicians track disease progression over time and assess the effectiveness of treatments. Speech is evaluated on a scale from 0 (Normal) to 4 (unintelligible most of the time), measuring the influence of PD on communication in everyday activities. The UPDRS measures speech deficits resulting from Parkinson's disease (PD) in two sections: Activities of Daily Living (ADL) and Motor Examination. In ADL, the scale ranges from 0 to 4 and is defined as follows.

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TABLE 1. Stages of parkinson's disease [10].

Category	Description
Early Stage (Stage 1)	Patients with Parkinson's disease find the least disturbance from regular responsibilities. A single side of the body shakes alone.
Moderate Stage (Stage-2)	Resting tremors, stiffness, and shaking might affect any side of the body. Furthermore, PD individuals' facial expressions could vary.
Mild Stage (Stage 3)	Mild stage-3 PD patients experience notable changes, including loss of stability and lower dexterity. Medication and occupational therapy might help lessen signs.
Progressive Stage (Stage 4)	This stage becomes worse, making walking without help, like walkers, complex.
Advanced Stage (Stage 5)	The most significant severe and developed level of Parkinson's disease. One might get hypothermia if they stand with rigid legs. Patients may suffer from hallucinations and delusions and find it challenging to remain motionless without collapsing.

- 1) **Normal:** No noticeable speech difficulties.
- 2) **Mildly affected:** The patient is understood without effort but may show subtle changes in speech.
- 3) **Moderately affected:** The patient is sometimes asked to repeat statements for clarity.
- 4) **Severely affected:** The patient is frequently asked to repeat statements, indicating significant speech issues.
- 5) **Unintelligible most of the time:** The patient's speech is difficult to understand for most listeners.

The Motor Examination assesses motor-related speech impairments, with scores from 0 (normal) to 4 (entirely incomprehensible), indicating problems such as monotone delivery, slurring, or diminished loudness due to motor control deficiencies. In stage classification for Parkinson's disease, UPDRS scores for speech provide quantifiable labels for machine learning models. These scores correspond to levels of disease severity of the multiclassification used in the proposed model: Health Control (UPDRS 0), early (UPDRS 1), moderate (UPDRS 2-3), and advanced (UPDRS 4). Speech recordings may be analyzed to correlate pitch fluctuation, loudness, and articulation with UPDRS scores. These evaluations underscore speech problems as significant indicators for Parkinson's disease, facilitating the categorization of disease development. Employing structured assessments with modern deep learning methodologies may facilitate multiclass categorization of Parkinson's disease stages, providing insights for individualized treatment and enhancing diagnostic precision [11].

Modern speech processing [12], [13], [14] technology constantly changes and mostly depends on multimodal signal processing and interdisciplinary research in Artificial Intelligence (AI). Many approaches [15], [16], [17] have recently been developed to solve different Human-Computer Interaction (HCI) problems. Considering these studies, several Machine Learning (ML) and Deep Learning (DL) models have been used for PD prediction. While most previous studies employed continuous speech features without considering whether the sequences of features correlated, most past studies concentrated [18], [19] on discovering relevant static features for PD speech classification. Existing research works employed DL and ML methodologies to improve Parkinson's disease prediction. Seventy to ninety percent of people with Parkinson's disease have dysphonia or speech

impairment. The primary warning sign is disorganization or modification in speech. In 2021, the diminished pitch, fast bursts, strained vocalizations, and extended pauses of people with PD made their speech difficult to comprehend. Parkinson's Disease occurs in men 1.5 times more often than in females, with around 10 million persons already diagnosed worldwide, and projections indicate an increase. Annually, around 60,000 individuals in the United States are diagnosed with Parkinson's disease, primarily impacting people over the age of 50. It is observed that there has been a 4% incidence rate of Parkinson's disease among individuals [20].

Bukhari and Ogudo [21] presented an ensemble machine learning model using the AdaBoost classifier to identify Parkinson's disease from speech signals, including SMOTE for data balance and PCA for dimensionality reduction. As a non-invasive, economical diagnostic instrument, it attains remarkable performance. Nonetheless, the model's computational complexity, the need for hyperparameter tuning, and dependence on a restricted dataset impede its wider clinical use, apart from the constraints related to the explainable AI component of the study. Recently, machine learning algorithms were used to categorize Parkinson's disease (PD) patients and Healthy Controls (HC) based on speech data, with the intention mainly of binary classification.

Abdel-Fattah et al. [22] employed a hybrid machine learning approach to advance the categorization of PD patients utilizing a dataset that included 756 speech measurements and achieved better performance. Integrating supplementary biomarkers, such as handwritten images, to establish a more comprehensive decision-making framework for diagnosing Parkinson's disease can be done in the future. Additionally, the integration of advanced swarm algorithms with inventive feature selection techniques has the potential to enhance classification performance and optimize results. Furthermore, Dao et al. [23] investigated vocal features for Parkinson's disease classification using a dataset of 756 samples and 754 features, underscoring the efficacy of speech signal analysis in differentiating Parkinson's patients from healthy subjects.

Existing research highlights ML and DL models' efficacy in binary categorization of Parkinson's disease using speech data. Nonetheless, studies about stratifying Parkinson's disease severity into several phases based on speech features

have a significant deficiency in the lack of transparency and issue of overfitting problem. Furthermore, rather than categorizing patients into two categories, a stage classification framework based on numerous severe Parkinson's disease symptoms should be created. Hence, we planned to propose a data-driven deep learning framework for stage categorization that may provide a more refined comprehension of disease development, delivering significant insights for patient surveillance and tailored treatment approaches.

Many research studies have used deep learning methods to investigate the use of speech data for PD prediction. Using a mix of Correlation Feature Selection (CFS) and Convolutional Neural Networks (CNNs) on audio samples, Costantini et al. [24] attained an accuracy of 61% for multi-classification tasks. However, there is a lack of transparency, and the dataset size is small. Fujita et al. [25] used Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) models were applied to speech data from 22 HC and 30 PD participants. An accuracy of 70% was obtained, but a small dataset was used for this study. Maskeliūnas et al. [26] presented a U-Net-like Deep Neural Network (DNN) technique with 89.64% and 79.49% accuracy, using Italian PVS and their Lithuanian PD speech dataset; however, it lacked feature selection and transparency. Gunduz [27] used a 9-layer CNN paired with a Support Vector Machine (SVM) with UCI speech data, showing an accuracy of 86.9% with Tunable Q-factor Wavelet Transform+Mel-Frequency Cepstral Coefficients concatenation (TQWT+MFCC+Concat features). This study performed binary classification but failed to address class imbalance issues. Notwithstanding these developments, most current studies concentrate on binary classification (PD vs. HC) and less on multiclassification for staging PD. This work fills this need using speech data to achieve multiclassification and differentiate early, mid, and advanced PD phases. Hence, we proposed a Parkinson's Speech Stage-Net (PSS-Net) model, which uses speech data for stage classification in PD phases, using a robust deep learning architecture and improving clinical confidence and application.

The main contributions of the proposed work are as follows:

- This work is to employ speech data to achieve stage classification for Parkinson's disease (PD), filling a gap in previous research works focusing on binary classification (PD vs. healthy controls).
- Establishment of a Resilient Deep Learning Framework: We propose a robust deep learning pipeline that uses advanced preprocessing techniques, such as Isolation Forest for outlier detection, KMeansSMOTE for class imbalance, Optimized Dimensionality Reduction (OptDimReduceX), and Random Forest-based feature selection (TreeSelect) method, to identify the most significant speech features.
- Stage Classification Accuracy: The Optimized and Fused Neural Network (OptiFusionNet) model attained an accuracy of 89% in categorizing Parkinson's disease into early, mid, and advanced phases. This effective stage classification

underscores the potential of speech-based models to aid in monitoring disease progression.

- Model Interpretability and Transparency: The study used Shapley Additive Explanations (SHAP) to ensure the model's interpretability and openness. This enhanced the model's reliability and therapeutic applicability since clinicians could understand how individual speech qualities contribute to classification.

- Comparative analysis with state-of-the-art techniques.

The rest of this work is organized as follows: Section II provides a literature review, summarizing relevant research works carried out and highlighting research needs. Section III discusses the methodology, which includes data preparation, feature selection, and the proposed deep learning architecture. Section IV discusses the findings, focusing on the model's performance and accuracy. Finally, Section V summarizes the study's essential results, contributions, and future research prospects.

II. RELATED WORKS

Deep Learning (DL) and machine learning techniques applied to speech data have shown potential in identifying PD, mainly by concentrating on binary classification (Parkinson's Disease(PD) vs. Healthy Control(HC)). Nonetheless, stage categorization, which categorizes Parkinson's Disease into early, mid, and advanced stages, remains little examined. To tackle the issue of data scarcity in deep learning applications, several recent studies have investigated novel algorithms designed for limited and unbalanced datasets. This section reviews related research on speech-based Parkinson's disease stage classification using deep learning, highlighting the deficiencies in existing stage classification initiatives.

Hoq et al. [28] proposed a hybrid model based on a Support Vector Machine (SVM) integration with a Principal Component Analysis (PCA) and a Sparse Autoencoder (SAE) for the identification of PD patients based on speech features. Equilibrated using the SMOTE method, the PD speech dataset produced an accuracy of 94.4%. However, there is a lack of different feature selection methods used. Rising an accurate Parkinson's disease prognosis system is essential, according to Soumaya et al. [29]. Together with speech acoustic and decomposition features for the Support Vector Machine classifier, the authors used an improved technique and evaluated the optimization approach of evolutionary algorithms. Liu et al. [30] put up a thorough ensemble approach. Local discriminant Preservation Projection (LPP) emphasizes PD prediction. Using an optimum strategy, the ensemble technique reduces the intra-class variation of the samples and preserves the neighborhood arrangement of PD speech samples concurrently. Unbalanced PD samples were under sufficient control. To lower dimensionality, feature selection should have been used to help lessen the curse of dimensionality.

Goyal et al. [31] assessed many classification techniques to show the effectiveness of every classifier. The Significance of Genetic Algorithm (GA) approaches, maximum

Relevance Minimum Redundancy (mRMR), and Principal Component Analysis (PCA) were also investigated to lower the cardinality of the speech dataset without significantly compromising accuracy by looking at three independent parameters. The XGBoost classifier, independent of class imbalance, guaranteed improved accuracy. The possible use of feature selection approaches is investigated regarding the usage of vocal aspects in Parkinson's disease. Using a unique feature set (40PD, 40 Healthy Control) and the MI-AE-GOLGBM technique, Dhar [32] improved the categorization of advanced Parkinson's disease and attained a performance rate of 84.17%. Future studies focused on including strategies to enhance categorization accuracy. Liu et al. [33] investigated patient remote monitoring using a 50-attribute feature set. This work used Shapley Additive explanations with a Light Gradient Boosting Machine, SHAP-LightGBM, showing a unique feature selection and modeling method. With a performance result of 91.62%, the conventional feature selection method was used. Lamba et al. [34] assessed the Parkinson's disease diagnosis accuracy of Naive Bayes, k-NN, and Random Forest (RF) techniques. This study presented a consistent feature extraction and selection technique incorporating a hybrid Mutual Information Ranking and Feature Elimination MIRFE-XGBoost. Although the model showed improved performance, there was a lack of transparency in PD prediction. Hawi et al. [35] assessed how rigorous speech treatment improved functional communication. Many studies have been undertaken with great effort to identify Parkinson's disease.

Alalayah et al. [36] presented a novel method using speech analysis to improve the strategies for early Parkinson's Disease detection. If features are inadvertently overlooked, excluded, or excessively displayed, biases might result. With high classification accuracy, the model effectively recognized PD symptoms. Iyer et al. [37] examined spectrograms generated from speech recordings of individuals diagnosed with Parkinson's disease using a pre-trained CNN (InceptionV3). With an AUC of 0.97 for colorful spectrograms and 0.96 for monochromatic spectrograms, the DL model improved performance in identifying individuals with Parkinson's disease. By using many ML techniques, including a three-layer Deep Neural Network2 (DNN2), which exceeds all other approaches with an accuracy of 95.41%, Rahman et al. [38] seek to identify Parkinson's disease in its early-stage PD obtained from speech signals. DNNs show improved efficiency compared to standard machine learning methods in patient classification.

Alshammri et al. [39] evaluated several ML models to create a soft diagnostic tool for Parkinson's disease prediction using speech signal features. Among the ML models, the MLP performed with an overall accuracy of 98%, in addition, with a remarkable accuracy of 95%, the model based on the Support Vector Machine (SVM) outperformed earlier machine learning models to identify people with PD. However, the input dataset is significantly

limited in size and binary classification is done. Saxena and Andrew [40] proposed a classification framework for Parkinson's disease diagnosis through machine learning models, including SVM, KNN, Random Forest, and Logistic Regression. This study achieved an accuracy of 89.47% on the testing set. Nevertheless, the suggested approach ignores explainable insights and class imbalance, essential to increase interpretability and confidence in medical diagnosis.

Chawla et al. [41] presented a nature-inspired technique using the Zebra Optimization Algorithm (ZOA) and Recursive Feature Elimination Cross-Validation (RFECV). This approach finds PD via feature selection based on a Novel Information Filter System (NIFS). Depending on the many features of the dataset, it provides the possibility to diagnose additional diseases. Singh and Tripathi [42] used machine learning techniques to identify speech-based features of Parkinson's disease, focusing on reducing computational complexity and improving accuracy. The Ensemble Feature Selection Algorithm (EFSA) technique achieved superior performance across three datasets. However, the study struggles to evaluate the severity stages of Parkinson's disease, a crucial element given the slow nature of the disease's course. Given the difficulty of concurrently training many models, this method depends on computer resources, which limits it. Furthermore, without using explainable AI and depending on the quality and amount of the training dataset may still determine the model's accuracy. Shyamala and Navamani [43] proposed Interpretable Feature Ranking XGBoost (IFRX) comprises a succession of procedures, including SVM SMOTE for class balancing, RFE with XGBoost for feature selection with classifier, and SHAP for explainability. Even though this work has shown improved performance, the input dataset is considerably small.

Pragadeeswaran and Kannimuthu [10] developed an automatic PD detection using the Adaptive Intelligent Polar Bear (AIPB) Optimization-Quantized Contempo Neural Network (QCNN) framework. This study's innovative approach improves classification accuracy but faces challenges like hyperparameter optimization and limited multi-label classification. Future improvements may focus on categorizing PD phases. Escalona et al. [44] introduced a Multi Condition Training (MCT) framework for Parkinson's disease detection, utilizing realistic acoustic stimuli and diadochokinetic speech tests. The MCT technique outperforms Single-Condition Training (SCT) with mean accuracy gains of 1.68%, 5.18%, and 4.39%, promoting further research in noise management. This innovative technique for noise management in PD detection inspires further study, notably in PD stage monitoring and stage classification under noisy situations employing speech recordings. Abumalloh et al. [45] highlighted the growing use of Deep Learning (DL) for identifying PD. Key findings include shortcomings in incremental learning for extensive data analysis and the need for more detailed reporting on datasets like demographic information. Rabie and Akhloufi [46] performed

an extensive assessment of machine learning and deep learning methodologies for the identification of Parkinson's disease using multimodal data, including speech, gait, and imaging. They emphasized significant constraints of data scarcity in healthcare, including limited dataset size, data asymmetry, and privacy issues, and examined methods such as augmentation (SMOTE, GANs), hybrid learning, and incremental models to enhance model robustness and generalizability.

Valarmathi et al. [47] introduced a Feature-Based Deep Neural Network (FB-DNN) that integrates Autoencoders and Deep Neural Networks (DNNs) for the identification of Parkinson's Disease (PD) using vocal inputs. To address limited and geographically constrained datasets, they used signal augmentation (Modified Band Pass Filter) and unsupervised feature extraction (SAE) to mitigate data scarcity, providing a cost-effective and scalable deep learning solution for telemedicine and early Parkinson's disease screening. Naeem et al. [48] investigated machine learning models for Parkinson's disease prediction using voice biomarkers and a limited dataset of 195 samples. They addressed data scarcity by using PCA for dimensionality reduction and SMOTE for class balancing, highlighting practical strategies such as feature optimization and model simplicity to get excellent performance (RF: 94% accuracy) in contexts with limited data.

Recent deep learning research has started the evaluation of Parkinson's disease (PD) severity staging using several data modalities. Ambulatory dynamics with wearable sensors: Aşuroğlu and Oğul used foot pressure (ground reaction force) sensors and a hybrid CNN/regressor to accurately predict UPDRS results. Their framework attained a correlation of 0.897 and 99.5% accuracy for binary Parkinson's disease identification [49]. They emphasized the constraints of their dataset, observing a limited PD population and an under-representation of severe patients with high-UPDRS scores. Navita et al. [50] suggested a dual-stage model using vertical force sensors: a Random Forest for classifying Parkinson's Disease vs control, and an ensemble regressor for assessing severity. Based on data from 93 Parkinson's disease (PD) patients and 73 controls, they claim a mean accuracy of 97.5% for PD identification and 96.4% for severity regression. Park et al. [14] integrated multimodal clinical, lifestyle, and wearable sensor data to delineate data-driven categories of Parkinson's disease severity. Their clustering analysis demonstrated that the main components of ankle gait signals serve as the most robust markers of severity: a classifier using these characteristics flawlessly distinguished between severe and moderate subtypes (AUC = 1.0). In conclusion, wearable sensor methodologies attain great precision by measuring gait characteristics; nevertheless, future research must broaden datasets and evaluate resilience in uncontrolled settings. Channa et al. [51] employed inertial sensors (Shimmer3 accelerometer) affixed to Parkinson's patients to record resting tremor throughout everyday activities. They retrieved statistical and spectral

tremor characteristics and used resampling to rectify class imbalance, assessing classifiers (XGBoost, decision tree, KNN). XGBoost with random oversampling had the greatest multi-class tremor-severity accuracy of 99% compared to decision tree and KNN. Results were presented according to severity level (0–(4)), rather than only as an aggregate detection. However, limitations include a restricted sample size, reliance on data from a single limb, and the need for validation using larger, independent datasets.

In summary, the research work analyzed revealed substantial progress in speech-based Parkinson's disease (PD) using Machine Learning (ML) and Deep Learning (DL) techniques. Methods such as feature selection, ensemble techniques, and innovative classifiers have enhanced binary classification precision and tackled class imbalance, dimensionality reduction, and feature significance. The results underscore a significant deficiency in stage classification, namely in categorizing Parkinson's Disease into early, mid, and advanced phases, which is crucial for tracking disease progression and customizing therapies. To overcome these constraints, we present the Parkinson's Speech Stage-Net model (PSS-Net), an intense deep learning architecture for stage categorization of Parkinson's disease stages using vocal data. PSS-Net seeks to boost accuracy and enable practical application in healthcare using sophisticated deep learning methods, emphasizing feature extraction and implementing stage-specific categorization.

III. METHODOLOGY

The methodology employed in this study presents a structured and systematic approach to classify Parkinson's Disease (PD) severity stages based on speech data. This approach integrates advanced preprocessing, feature engineering, and model development. Critical stages of the section include data preprocessing with outlier identification and normalization, class balancing using KMeansSMOTE, and feature selection using the tree-based model. An optimized and Fused Neural Network (OptiFusionNet) combines Dropout and Batch Normalization to ensure accuracy and resilience. This end-to-end system presents a novel application of existing methods, offering PD severity categorization as an efficient approach.

A. OVERVIEW

The proposed model pipeline begins with importing the dataset and cleaning it by deleting the ID and Gender columns, which are unimportant. We use Min-Max scaling to normalize relevant speech components related to PD.

The relevant speech features were chosen based on previous studies indicating their significance in the course of Parkinson's Disease. These include Jitter, Shimmer, HNR, MDVP parameters, RPDE, NHR, Spread1, Spread2, D2, PPE, and DFA. We assigned weights to these features to represent their respective impacts on the severity index under our proposed framework. The composite index was used to categorize individuals into four classifications: healthy, mild, moderate, and severe based on quantile criteria derived

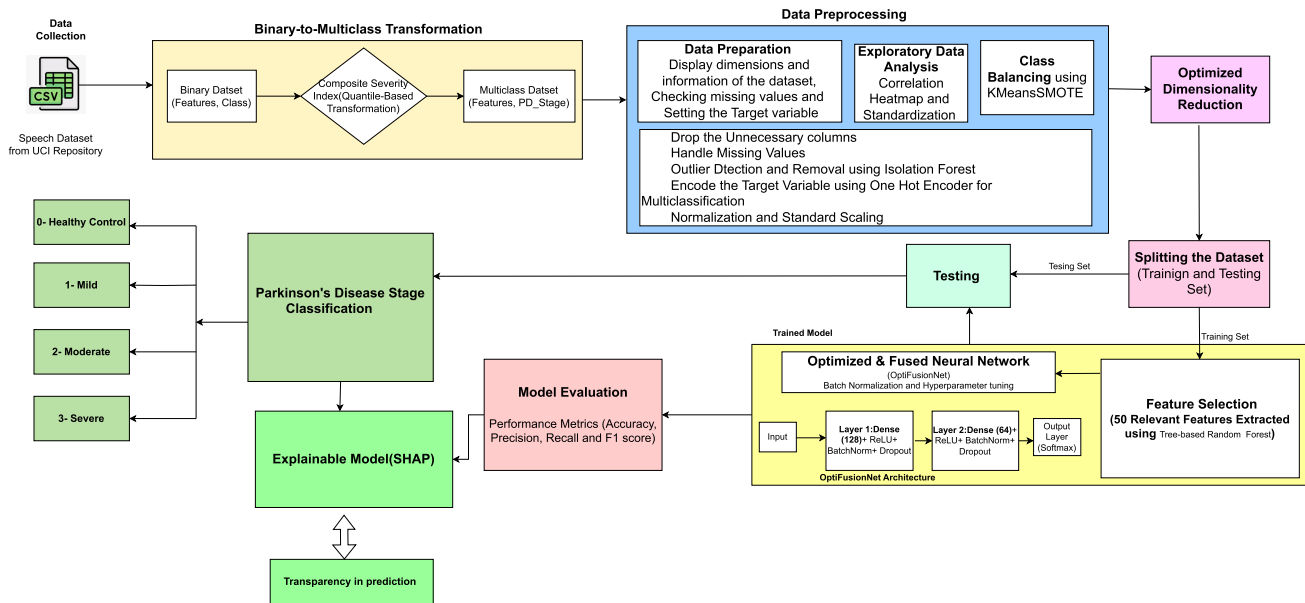


FIGURE 1. Overview of the proposed model.

TABLE 2. Essential speech features for parkinson's disease multiclassification.

Feature Name	Technical Specification	Description	Data Type
DFA	Detrended Fluctuation Analysis	Measures signal complexity, indicating fractal properties of speech dynamics.	Float64
PPE	Pitch Period Entropy	Quantifies the variability in pitch, reflecting monotonicity in speech.	Float64
RPDE	Recurrence Period Density Entropy	Captures non-linear vocal patterns and helps detect speech irregularities in PD.	Float64
std_delta_log_energy	Standard Deviation of Log Energy Delta	Measures energy variation, highlighting changes in speech dynamics.	Float64
mean_MFCC_1st_coef	Mean MFCC 1st Coefficient	Encodes spectral features of speech, capturing vocal quality changes.	Float64
locPctJitter	Local Percentage Jitter	Reflects irregularities in vocal fold vibrations, often seen in Parkinson's speech.	Float64
locShimmer	Local Shimmer	Indicates amplitude variation, highlighting vocal instability.	Float64
meanAutoCorrHarmonicity	Mean Autocorrelation Harmonicity	Measures periodicity in the voice signal, often reduced in PD patients.	Float64
meanNoiseToHarmHarmonicity	Noise-to-Harmonics Ratio	Quantifies the voice's noise proportion, which is higher in Parkinson's speech.	Float64
numPulses	Number of Glottal Pulses	Total number of glottal pulses, indicating rhythm and stability.	Integer
stdDevPeriodPulses	Standard Deviation of Glottal Pulse Period	Measures variability in glottal pulse durations, indicating vocal instability.	Float64
GNE_mean	Glottal-to-Noise Excitation Mean	Indicates the quality of glottal excitation, which is altered in Parkinson's speech.	Float64
tqwt_energy_dec_18	Tunable Q-factor Wavelet Transform (TQWT) Energy at the 18th decomposition level	Measures the signal energy at the 18th wavelet decomposition level, helping detect dominant frequency components.	Float64
tqwt_entropy_log_dec_16	Tunable Q-factor Wavelet Transform (TQWT) logarithmic at the 16th decomposition level	Measures the logarithmic entropy at the 16th level, capturing the complexity or randomness of the signal.	Float64

from the distribution of severity scores. This process leads to the design of a data-driven framework for classifying PD

severity stages using a statistical approach. The labels are given according to certain conditions using the initial severity

index values and the binary classification (class). The final dataset is ready for analysis, with weighted features (X) passing through and multiclass labels (y) coming out.

Figure 1 illustrates a systematic process for developing a deep learning-based system for classifying Parkinson's disease. Initially, the speech dataset is transformed from binary to multiclass, followed by robust data preprocessing, including class balancing, normalization, Dimensionality reduction, and standardization. The dataset is divided into training and testing, and essential features are extracted to improve model accuracy. Machine learning models are used in model training and optimization, including Logistic Regression, Random Forest, Gradient Boosting, SVC, XGBoost, and LightGBM. Deep learning models include CNN, MLP, and Artificial Neural Network. Batch normalization improves convergence for deep learning models. Grid Search CV is used in hyperparameter tuning to maximize early stopping, batch size, learning rate, and dropout rate. Accuracy, precision, recall, and F1 score are used in the model assessment. Explainability is used in SHAP to understand feature contributions. This approach guarantees stage categorization that is reliable, comprehensible, and effective. The proposed Parkinson's Speech Stage-Net (PSS-Net) model makes Reliable Parkinson's disease severity classification possible, emphasizing model improvement and assessment. Finally, explainable AI (SHAP) integration provides transparent and interpretable outcomes. The functionality of each phase is mentioned, and the proposed model showed better performance.

B. DATASET DESCRIPTION

The proposed model was assessed using the benchmark dataset and the severity of Parkinson's disease by the University of California Irvine (UCI) Machine Learning Repository [52]. Table 2 highlights the dataset's key speech features, technical details, descriptions, and data formats. These features are essential for identifying patterns and inconsistencies in speech linked to Parkinson's disease, which helps with stage categorization. These features collectively capture the key dimensions of speech affected by Parkinson's disease, such as Pitch stability, amplitude fluctuation, noise levels, non-linear dynamics, and energy changes. The peculiarities of Parkinson's speech are robustly represented when these features are used for stage categorization.

Dataset Description adds additional phases of Parkinson's disease (mild, moderate, severe) to an existing binary dataset, transforming it into a multiclassification dataset. To convert a binary classification dataset into a stage classification dataset for the staging of Parkinson's disease (PD), we utilized a quantile percentile statistics approach. This methodology entails categorizing continuous speech-related features into distinct classifications corresponding to the various phases of PD.

The mathematical analysis of the computation of the PD severity index is as follows.

- 1) **Normalization:** For each feature f , the normalization is performed using the min-max scaling formula:

$$x_{j,f}^{(norm)} = \frac{x_{j,f} - \min(x_{:,f})}{\max(x_{:,f}) - \min(x_{:,f})}. \quad (1)$$

Where:

- $x_{j,f}$ is the original value of feature f for the j -th data sample.
- $x_{j,f}^{(norm)}$ is the normalized value of feature f for the j -th data sample.
- j represents the index of a specific data sample (or instance) in the dataset.
- f represents the index of a specific feature (or column) in the dataset.
- x represents the dataset or data matrix, where each row corresponds to a data sample and each column corresponds to a feature.

- 2) **Weighting:** Apply weights to normalized features:

$$x_{j,f}^{(weighted)} = w_f \cdot x_{j,f}^{(norm)}. \quad (2)$$

- 3) **Composite Index Calculation:** Compute the severity index:

$$S_j = \sum_{f \in \text{selected features}} w_f \cdot x_{j,f}^{(norm)}. \quad (3)$$

- 4) **Thresholding:** Define thresholds $Q_{0.33}$ and $Q_{0.66}$ from the distribution of S for PD cases.

- 5) **Stage Assignment:** Assign stages based on:

$$PD_Stage_j = \begin{cases} \text{Healthy Control} & \text{if } class_j = 0, \\ \text{Mild} & \text{if } class_j = 1 \text{ and } S_j \leq Q_{0.33}, \\ \text{Moderate} & \text{if } class_j = 1 \text{ and } Q_{0.33} < S_j \leq Q_{0.66}, \\ \text{Severe} & \text{if } class_j = 1 \text{ and } S_j > Q_{0.66}. \end{cases} \quad (4)$$

By employing quantiles, the dataset was classified into early, moderate, and advanced stages by establishing thresholds determined by percentile values. This transformation ensures the preservation of the data distribution while offering a methodical approach to stratifying patients according to the severity of their symptoms. By utilizing the UPDRS speech evaluation scores and implementing quantile-based thresholds, the binary classification labels (PD versus healthy controls) are effectively transformed into multiclass labels. This methodology augments the granularity of disease analysis, facilitating more accurate staging and customized therapeutic approaches.

Tables 3 and 4 summarize the demographic characteristics of the dataset utilized in this work. The ID attribute ranges from 0 to 251 and provides unique IDs for the 756 data samples. The Gender attribute shows the distribution of male and female participants in the dataset. The study is ensured to represent genders since the gender data exhibit a balanced distribution with no missing values.

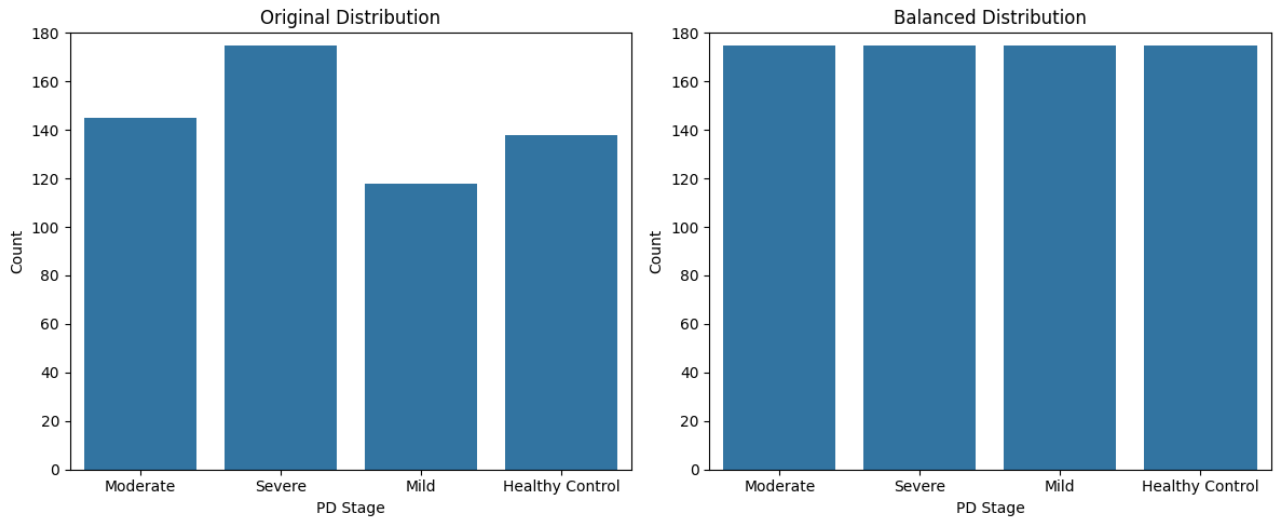


FIGURE 2. Comparison of class distribution before (Left) and After (Right) applying KMeansSMOTE for balancing Parkinson's disease severity stages.

TABLE 3. Demographic information of the dataset.

Attribute	Count	Mean	Std Dev	Min	25%	Median	75%	Max
ID	756	125.5	72.79	0	62.75	125.5	188.25	251
Gender	756	0.5159	0.5001	0	0	1	1	1

TABLE 4. Gender distribution in the dataset.

Gender	Count	Percentage
Male (1)	390	51.59%
Female (0)	366	48.41%
Total	756	100%

TABLE 5. Demographic information of the PD stages dataset.

PD Stage	Total	Female <i>n</i> (%)	Male <i>n</i> (%)
Healthy Control	192	123 (64.06%)	69 (35.94%)
Mild	148	80 (54.05%)	68 (45.95%)
Moderate	193	71 (36.79%)	122 (63.21%)
Severe	223	92 (41.26%)	131 (58.74%)
Overall	756	366 (48.41%)	390 (51.59%)

Table 5 shows the gender distribution across different Parkinson's Disease (PD) stages in a total of 756 subjects. Female representation is higher in the Healthy and Mild stages, while males are predominant in the Moderate and Severe stages.

C. PREPROCESSING

This phase is an essential component that the dataset should undergo a preprocessing step to clean and remove the noise in the speech data. Preprocessing was done on a pandas DataFrame imported from the dataset. First, non-relevant fields were removed to concentrate only on factors directly affecting PD severity categorization and extracted as the result variable was PD_Stage, the PD severity

levels. The ground truth for the classification model is defined from PD_Stage, guaranteeing that the approach followed the multiclass categorization. Label encoding and target encoding transform categorical variables into numerical representations for model compatibility, while outlier identification and removal improve data quality by detecting and eliminating anomalous values and are described mathematically as follows.

1. TARGET ENCODING

Label encoding converts categorical labels into integer codes. Each unique category is assigned a distinct integer.

$$y_i^{(\text{encoded})} = f(y_i) \quad (5)$$

Where:

- y_i : The original target label for the i^{th} observation.
- $f(\cdot)$: The label encoding function maps each category to a unique integer.
- $y_i^{(\text{encoded})}$: The encoded numerical representation of the original label.

2. ONE-HOT ENCODING

One-hot encoding converts the integer labels into binary vectors, where each class is represented by a vector with a 1 in the position corresponding to the class and 0 elsewhere.

$$y_i = (\delta(y_i^{(\text{encoded})}, 0), \dots, \delta(y_i^{(\text{encoded})}, K - 1)) \quad (10)$$

Where:

- y_i : The one-hot encoded vector for the i^{th} observation.
- $y_i^{(\text{encoded})}$: The integer label from label encoding.
- K : The target variable's total number of classes (categories).
- $\delta(a, b)$: The Kronecker delta function, defined as:

$$\delta(a, b) = \begin{cases} 1 & \text{if } a = b \\ 0 & \text{if } a \neq b \end{cases}$$

Algorithm 1 Workflow for PD Severity Classification Using KMeans-SMOTE

- 1: **Start:** Initiate the process.
- 2: **Load Speech Dataset:** Load dataset D with imbalanced classes: Healthy Control (HC), Mild (M), Moderate (Mo), Severe (S).
- 3: **Preprocessing:** Normalize and standardize features.
- 4: **Apply K-Means Clustering:** For each minority class $C_i \in \{HC, M, Mo, S\}$, perform K-means clustering:

$$C_i = \{c_{i1}, c_{i2}, \dots, c_{iK}\} \quad (6)$$

where K is the number of clusters.

- 5: **Cluster Classes:** Divide each class C_i into K clusters.
- 6: **for** each cluster c_{ij} in C_i **do**
- 7: **Apply SMOTE:** Generate synthetic samples S_{ij} within c_{ij} :

$$S_{ij} = SMOTE(c_{ij}) \quad (7)$$

- 8: **end for**
- 9: **Check Balancing Ratio:** Verify if the desired balancing ratio is achieved:

$$BR = \frac{|HC| + |M| + |Mo| + |S|}{4} \quad (8)$$

- 10: **if** balancing ratio not achieved **then**
- 11: Repeat SMOTE generation.
- 12: **else**
- 13: Proceed.
- 14: **end if**
- 15: **Combine Data:** Merge synthetic samples with the original dataset:

$$D_{final} = D_{original} \cup \bigcup_{i,j} S_{ij} \quad (9)$$

- 16: **Train Model:** Train a classifier to distinguish between HC, M, Mo, and S.
- 17: **Output Predictions:** Classify each sample as one of the PD severity levels.
- 18: **End** Stop the process.=0

3. OUTLIER DETECTION AND REMOVAL

Outliers can be detected using methods like Isolation Forest, which identifies data points significantly different from the majority.

a: A. OUTLIER PREDICTION USING ISOLATION FOREST

$$\text{outlier_prediction}_i = \begin{cases} 1, & \text{if normal} \\ -1, & \text{if outlier} \end{cases} \quad (11)$$

where:

- $\text{outlier_prediction}_i$: Indicates whether the i^{th} observation is normal (1) or an outlier (-1).
- $\alpha = 0.05$: The contamination parameter represents the proportion of outliers in the dataset.

LabelEncoder converted the category target variable (PD_Stage) into numerical form, thus preparing the data for model training. The encoded labels were then one-hot encoded to translate them into a format suitable for stage classification. Instead of a single scalar value, this method enables the model to predict the probability distribution across multiple classes. StandardScaler was used for feature normalization, standardizing the feature data to provide a mean of 0 and a standard deviation of 1. This normalization ensured that every feature contributed equally to the model training, reducing the impacts of different sizes or magnitudes in the feature collection. The Isolation Forest technique for outlier detection helped systematically identify and eliminate anomalous data points from the dataset. It separates anomalies using random partitions in the feature space, identifying spots needing fewer to be isolated and setting the contamination parameter to 0.05 anticipated outliers in around 5% of the dataset. Eliminating data points labeled as anomalies produced a cleaner dataset with better model generalizability. Following the preprocessing phase, it is imperative to assess the preprocessed data to determine whether the class distribution is balanced. The dataset is imbalanced, and to solve this issue, we used the KMeans-SMOTE technique to balance the classes. This technique for class balancing has improved the performance and outcomes of the model compared to traditional sampling techniques like SMOTE.

1) CLASS BALANCING

KMeansSMOTE effectively addresses class balancing when converting the binary classification dataset into stage classification problems. By creating synthetic samples for under-represented classes, this method equalizes the proportion of participants across early, moderate, and advanced phases of Parkinson's disease, therefore addressing unbalanced classes. It creates synthetic samples that preserve class borders and represent the natural feature distribution within each class by adding KMeans clustering, as shown in Algorithm 1. KMeansSMOTE involves applying the KMeans algorithm on the minority samples after selecting a predetermined number of clusters (k). The speech dataset's minority class occurrences are grouped into k groups using clustering. For each cluster formed inside the minority class, KMeans-SMOTE generates synthetic samples. The following phases are involved in the process: selecting a random sample from the cluster, identifying its k -nearest neighbors within the same cluster, and creating synthetic samples by interpolating between the neighbors and the chosen sample. By taking into account the local structure of the data, it produces a wider range of synthetic samples, improving the caliber of the generated instances [53]. As shown in Figure 2, this ensures a more balanced classification and improves the model's generalization.

KMeansSMOTE successfully preserves the correlations among important speech features for speech-related PD

datasets to ensure the synthetic data aligns with the original distribution. Applying KMeansSMOTE after the quantile-based transformation improves classification accuracy and ensures that minority classes are adequately represented, contributing to a stronger and more clinically applicable staging framework. Nevertheless, a thorough review is required to prevent the production of unrealistic or noisy samples that can impact the model's performance. The KMeans Synthetic Minority Oversampling Technique (KMeansSMOTE) solved a common problem in stage classification, as shown in Figure 2. This ensured a fair distribution of classes, thus enabling the model to learn efficiently at all PD intensity phases and lowering the danger of bias toward majority classes.

D. OPTIMIZED DIMENSIONALITY REDUCTION(OPTDIMREDUCEX)

As the dataset features are highly correlated, Principal component analysis was incorporated and optimized to decrease dimensionality, simplify computations, and eliminate feature space redundancy. PCA retained 95% of the variance and reduced the feature set into a lower-dimensional space. This dimensionality-lowering method improved model efficiency without compromising important data for precise categorization, as shown in Figure 3. The X-axis (Principal Component 1) depicts the direction with the most significant variation in the dataset, capturing the data's most important patterns. The Y-axis (Principal Component 2) is the second most significant direction, representing variance orthogonal to the first component. These two main components work together to give a lower-dimensional display of high-dimensional data, allowing one to discover patterns, clusters, and trends while retaining as much information as possible. This makes PCA an effective technique for exploratory data analysis and dimensionality reduction.

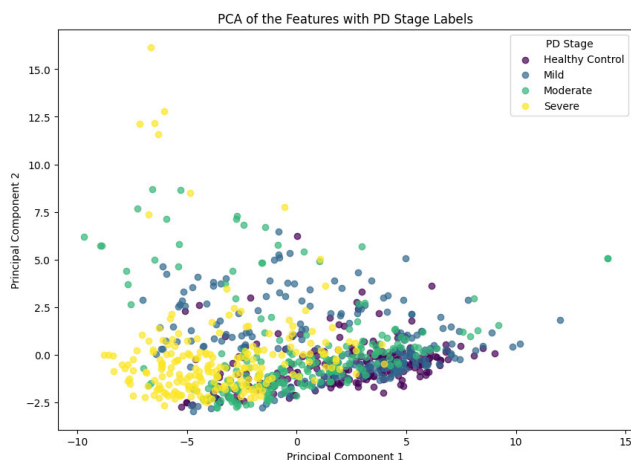


FIGURE 3. PCA-based optimized dimensionality reduction.

We implemented an optimized dimensionality reduction technique in our proposed model, which performs better than conventional linear dimensionality reduction methods. The proposed solution deliberately addressed the distinctive

features of speech-related neurological data, in contrast to default PCA algorithms that use general variance preservation. We designed a precision-driven dimensionality reduction technique that maintains important speech feature information. Also, improving computing efficiency by decisively keeping 95% of the dataset's variance. The Optimized Dimensionality Reduction (OptDimReduceX) approach focuses on developing a more condensed yet informative feature space unique to speech patterns associated with Parkinson's disease. The variance retention threshold was carefully chosen to preserve the subtle speech signal properties necessary for precise disease severity classification while reducing dimensionality. This method enabled us to improve feature representation, minimize computational complexity, and delete duplicate or less valuable features while maintaining the dataset's potential. The implementation is a domain-specific adaption of PCA, changing a standard statistical approach into a focused tool for analyzing neurological speech data. Thus, it boosts the robustness and efficiency of the proposed deep learning classification system.

E. FEATURE SELECTION (TREESELECT)

A tree-based Random Forest (TreeSelect) classifier was used for feature selection as it offers a metric of feature relevance based on its influence on the model's performance. Training on a balanced dataset, the Random Forest model assessed the value of every feature. The top 50 items with the highest significance ratings were selected for further study. Visualizing the feature significance using a bar plot revealed the most significant elements influencing PD severity categorization, stressing the value of the chosen speech features.

The feature significance scores reflect how important each feature is to the tree-based Random Forest model's decision-making process. The higher the score, the more significant the trait in predicting the severity of PD. We focus on the most important ones by identifying the most impactful speech features that distinguish severity levels of PD. This helps to optimize the feature set and improves the model's performance. Figure 4 depicts the feature significance scores from a Random Forest model for predicting Parkinson's disease severity using speech data. The most significant features include energy-related metrics, MFCC-based features, and entropy measurements. The most critical elements were detected using a tree-based Random Forest(TreeSelect) model, including DFA (Detrended Fluctuation Analysis), f1, and PPE, effectively capturing non-linear temporal dynamics in speech data and detecting motor deficits impacting vocal control. Furthermore, variables obtained from the Tunable Q-factor Wavelet Transform (TQWT), such as energy, entropy, and standard deviation, were revealed as critical for identifying minor differences in speech patterns associated with various phases of Parkinson's disease. These selected features were pivotal in improving the performance of the proposed Parkinson's Speech Stage-Net model (PSS-Net),

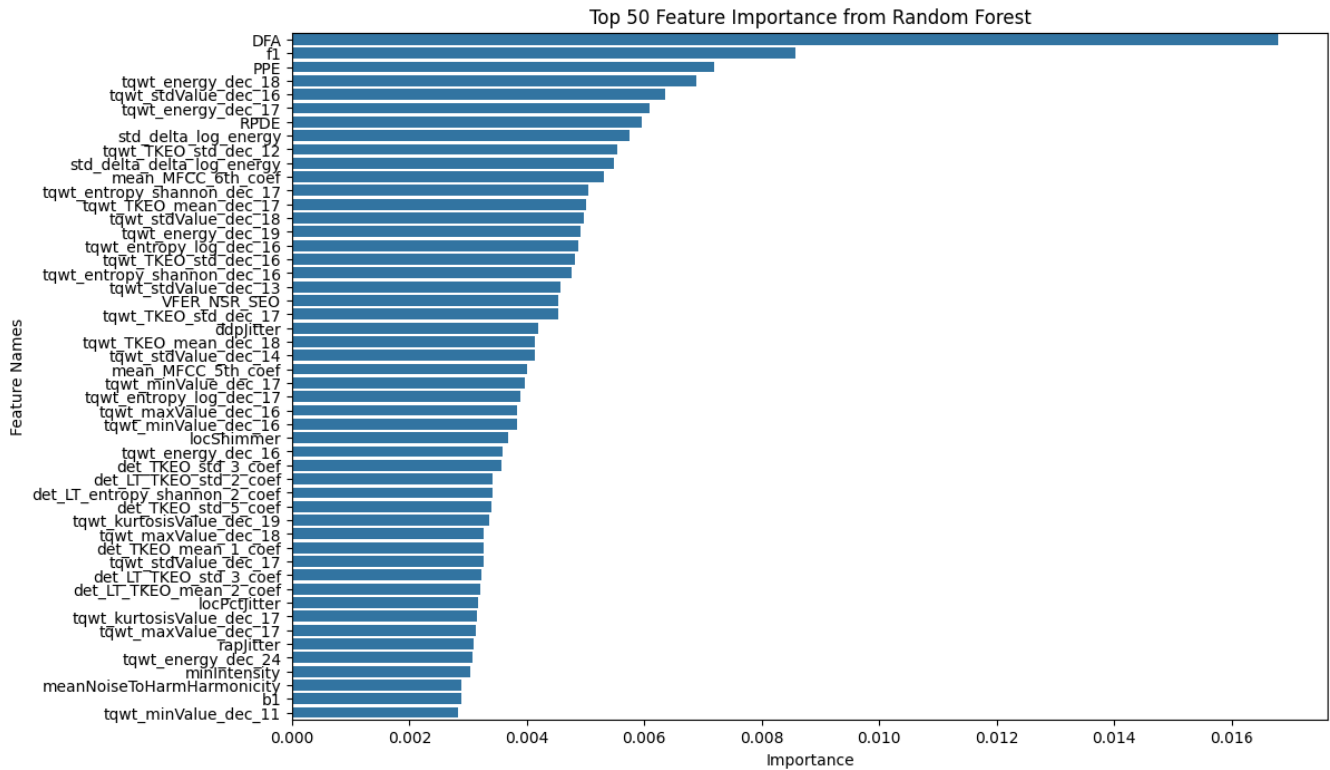


FIGURE 4. Essential features for parkinson's disease severity classification using tree-based random forest model.

allowing for reliable stage classification of PD stages and filling the research gap in stage-specific techniques.

F. MODEL TRAINING AND HYPERPARAMETER TUNING

During the model training and assessment phase, machine learning and deep learning classifiers are used on a pre-processed and balanced dataset to categorize PD severity occurrences, as described in Algorithm 2. The processed dataset is split into training and testing subsets in an 80-20 ratio. Model development was done using the training subset, and performance assessment was done using the testing subset. This division impartially assessed the model's ability to generalize to inputs that cannot be processed. The employed classifiers comprise Logistic Regression, Random Forest, Support Vector Classifier (SVC), Multi-layer Perceptron (MLP), CNN [54], XGBoost, Gradient Boosting, and LightGBM [53]. The OptiFusionNet can be described mathematically as follows.

Assume an input vector $\mathbf{x} \in \mathbb{R}^d$ where $d = X_{\text{train}}.\text{shape}[1]$. The network comprises several layers:

1) LAYER 1: DENSE + RELU

The first layer is a dense (fully-connected) layer with 128 neurons. Let

$$\mathbf{W}^{(1)} \in \mathbb{R}^{128 \times d} \quad \text{and} \quad \mathbf{b}^{(1)} \in \mathbb{R}^{128}$$

be the weight matrix and bias vector, respectively. The pre-activation output is:

$$\mathbf{z}^{(1)} = \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}. \quad (12)$$

The ReLU activation function is applied element-wise:

$$\mathbf{a}^{(1)} = \text{ReLU}(\mathbf{z}^{(1)}) = \max(0, \mathbf{z}^{(1)}). \quad (13)$$

2) LAYER 2: BATCH NORMALIZATION

Batch normalization normalizes the activations using the statistics computed over the mini-batch. For each neuron i in the layer, the normalized activation is:

$$\hat{a}_i^{(1)} = \frac{a_i^{(1)} - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad (14)$$

where μ_B and σ_B^2 are the mini-batch mean and variance, and ϵ is a small constant for numerical stability. The layer then scales and shifts the normalized activation:

$$\tilde{a}_i^{(1)} = \gamma_i \hat{a}_i^{(1)} + \beta_i, \quad (15)$$

with learnable parameters γ_i and β_i .

3) LAYER 3: DROPOUT

Dropout randomly sets a fraction of the activations to zero with probability $p = 0.3$ during training. Let $\mathbf{r} \in \{0, 1\}^{128}$ be a random mask vector with:

$$r_i \sim \text{Bernoulli}(1 - p). \quad (16)$$

Algorithm 2 Algorithm 2: PSS-Net Model Pipeline for Stage Classification**0: 1. Dataset Conversion Using Quantile Percentile:**

0: Apply statistical quantile percentile techniques to convert the dataset into a suitable format for stage classification.

0: 2. Preprocessing:

0: **Scaling and Normalization:** Scale features to a uniform range and normalize them.

0: **Handling Missing Values:** Identify and impute missing values using mean, median, or mode.

0: **Outlier Removal:** Use Isolation Forest to detect and remove outliers.

0: 3. Feature Correlation Analysis:

0: **Correlation Heatmap:** Identify multicollinearity using a heatmap.

0: **Optimized Dimensionality Reduction:** Apply PCA to retain 95% of the variance.

0: 4. Target Encoding:

0: Use one-hot encoding for categorical target variables.

0: 5. Class Balancing:

0: Apply KMeansSMOTE to handle class imbalance.

0: 6. Dataset Splitting:

0: Split the dataset into 80% training and 20% testing subsets.

0: 7. Feature Selection:

0: Use Random Forest (TreeSelect) to calculate feature importance.

0: Select top features based on importance scores.

0: 8. Model Training:

0: **ML Models:** Logistic Regression, Random Forest, Gradient Boosting, SVC, XGBoost, LightGBM.

0: **DL Models:** CNN, MLP, OptiFusionNet with Batch Normalization.

0: OptiFusionNet consistently produces better accuracy.

0: 9. Hyperparameter Tuning:

0: Optimize models using GridSearchCV.

0: Adjust learning rate, batch size, verbosity.

0: Apply early stopping and learning rate scheduler.

0: 10. Model Evaluation:

0: Evaluate using Accuracy, Precision, Recall, F1 Score.

0: 11. Multiclass Prediction:

0: Predict multiclass labels on the test dataset.

0: 12. Explainability Using XAI (SHAP):

0: Use SHAP for model interpretability.

0: Analyze feature contributions and visualize importance.

0: 13. Output:

0: Generate final report with:

0: - Performance metrics

0: - Feature importance rankings

0: - SHAP visualizations

0: - Highlight better-performing model (OptiFusionNet)

End of Algorithm=0

The output after dropout is:

$$\mathbf{d}^{(1)} = \tilde{\mathbf{a}}^{(1)} \odot \mathbf{r}, \quad (17)$$

where $\odot$ denotes element-wise multiplication.

4) LAYER 4: SECOND DENSE LAYER + ReLU

The next dense layer has 64 neurons. Let

$$\mathbf{W}^{(2)} \in \mathbb{R}^{64 \times 128} \quad \text{and} \quad \mathbf{b}^{(2)} \in \mathbb{R}^{64}.$$

The pre-activation output is:

$$\mathbf{z}^{(2)} = \mathbf{W}^{(2)} \mathbf{d}^{(1)} + \mathbf{b}^{(2)}, \quad (18)$$

with activation:

$$\mathbf{a}^{(2)} = \text{ReLU}(\mathbf{z}^{(2)}). \quad (19)$$

This is again followed by Batch Normalization and Dropout (with the same mechanism as above).

5) OUTPUT LAYER

The final dense layer produces the output for stage classification. Let the number of classes be

$$K = y_{\text{train}}.\text{shape}[1].$$

TABLE 6. Hyperparameter configuration for the proposed model.

Hyperparameter	Value
Neurons per Layer	128 (Layer 1), 64 (Layer 2)
Activation Function	ReLU (Hidden), Softmax (Output)
Dropout Rate	0.3
Optimizer	Adam
Loss Function	Categorical Cross-Entropy
Learning Rate	Adjusted via ReduceLROnPlateau (factor=0.1, patience=5, min_lr=1e-6)
Early Stopping	Enabled (patience=10, monitor='val_loss')
Batch Size	32
Epochs	Up to 100 (stops early if no improvement)

With weights

$$\mathbf{W}^{(3)} \in \mathbb{R}^{K \times 64} \quad \text{and biases} \quad \mathbf{b}^{(3)} \in \mathbb{R}^K,$$

the logits are computed as:

$$\mathbf{z}^{(3)} = \mathbf{W}^{(3)} \mathbf{d}^{(2)} + \mathbf{b}^{(3)}. \quad (20)$$

The softmax activation function is applied to obtain probabilities:

$$\text{softmax}(\mathbf{z}^{(3)})_i = \frac{e^{z_i^{(3)}}}{\sum_{j=1}^K e^{z_j^{(3)}}}, \quad i = 1, \dots, K. \quad (21)$$

Model Compilation and Training

The model is compiled using the Adam optimizer and the categorical cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^K y_i \log(\hat{y}_i), \quad (22)$$

where y_i is the true one-hot encoded label and $\hat{y}_i$ is the predicted probability. Also, early stopping and learning rate scheduler hyperparameter configurations were done accordingly.

The OptiFusionNet architecture, based on deep learning, was intended to categorize PD severity stages. The OptiFusionNet model incorporates various optimized techniques to improve prediction performance and prevent overfitting. It consists of multiple dense layers, starting with a 128-neuron layer followed by Batch Normalization to stabilize learning and Dropout (0.3) to reduce overfitting. The second hidden layer has 64 neurons, followed by batch normalization and dropout for better generalization. The final output layer uses softmax activation, making it suitable for stage classification. The model is compiled with the Adam optimizer and categorical cross-entropy loss, ensuring efficient learning. Additionally, Early Stopping is implemented to halt training if the validation loss stops improving. At the same time, a learning rate scheduler (ReduceLROnPlateau) dynamically

adjusts the learning rate to enhance convergence, as shown in Table 6.

This approach ensures robust learning and improved accuracy while minimizing overfitting. ReLU activation of each layer introduced non-linearity and a data complexity capture mechanism to capture intricate patterns. Batch normalization layers followed every hidden layer to stabilize the training process and raise convergence. After each hidden layer, dropout layers randomly turn off a percentage of neurons during training, preventing overfitting. A probability distribution across the PD severity levels was generated using a softmax-activated output layer, facilitating multiclassification. The model was built using the Adam optimizer, a generally embraced deep learning optimization method, with categorical cross-entropy as the loss function. Here, we considered accuracy as the primary assessment tool for better prediction of PD severity levels.

G. MODEL ASSESSMENT

The trained model was tested on a subset to evaluate its generalizing performance. Using a training-testing split, the dataset is split into 80% for training and 20% for testing. Each model is trained using the training set and evaluated on the testing set. The PSS-Net deep learning model implements a Sequential neural network architecture with Dense, Dropout, and Batch Normalization layers. Early stopping and learning rate reduction techniques are applied to optimize training. For every level of PD severity, metrics including accuracy, precision, recall, and F1 score were calculated to comprehensively assess the model's classification capacity. A confusion matrix was produced to see how well the model predicted every class. Equations (25) to (28) delineate the methodology for calculating classified instances. Accurately categorized cases are denoted as TP (True Positive) and TN (True Negative). Conversely, misclassified occurrences are denoted by FP (False Positive) and FN (False Negative). Each class also computed Area Under the Curve (AUC) scores and Receiver Operating Characteristic (ROC) curves to show the model's discriminative capacity. These graphs were designed to show how well the model could separate PD severity phases [23].

$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} \quad (23)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (24)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (25)$$

$$F1 \text{ score} = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \quad (26)$$

H. EXPLAINABLE ARTIFICIAL INTELLIGENCE

eXplainable Artificial Intelligence (XAI) techniques such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) elucidate the predictions of ML and DL models. LIME augments

TABLE 7. Classification report of the proposed model.

Class	Precision	Recall	F1-score	Support
0	0.90	0.79	0.84	48
1	0.86	0.88	0.87	42
2	0.85	0.92	0.89	38
3	0.93	0.98	0.96	44
Accuracy			0.89	172
Macro Avg	0.89	0.89	0.89	172
Weighted Avg	0.89	0.89	0.89	172

interpretability by offering localized explanations for the predictions generated by the models. LIME facilitates the identification of the factors that most significantly influence the predictions of specific cases, providing clarity into the model's decision-making process on a case-by-case basis. SHAP is based on a notion from game theory known as Shapley values. SHAP assigns each feature an importance value for a particular prediction based on the Shapley values concept from cooperative game theory. These SHAP values help understand each feature's impact on the model's output, enhancing interpretability. SHAP values were generated to improve model interpretability. SHAP thoroughly analyzed how every attribute affected the model's predictions, illuminating the underlying neural network decision-making process. This stage helped explain the findings and addressed the opaque models of deep learning models [43].

Conclusively, the Parkinson's Speech Stage-Net model (PSS-Net) model pipeline integrates sophisticated statistical, machine learning, and deep learning approaches to provide a systematic and optimal methodology for stage categorization, as shown in Algorithm 2. This workflow ensures a well-structured, data-driven approach to stage classification by combining feature engineering, model selection, and optimization strategies to improve prediction performance.

IV. RESULTS AND DISCUSSION

The performance metrics of machine learning and deep learning models assessed for Parkinson's disease (PD) severity classification are presented in this section. Convolutional Neural Network (CNN), Logistic Regression, Random Forest, Gradient Boosting, Support Vector Classifier (SVC), eXtreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Multi-Layer Perceptron (MLP) and OptiFusionNet with batch normalization are the models considered. This work evaluated a model that classifies PD severity into four categories: Healthy Control, Mild, Moderate, and Severe. Our goal in assessing these models is to determine the most accurate and trustworthy method for classifying the severity of PD, which may be crucial for early intervention and clinical decision-making.

A. EXPERIMENTAL SETUP

The experimental framework for PSS-Net is designed for stage classification tasks, encompassing the computational

TABLE 8. Comparative analysis of baseline ML/DL models and the proposed model for PD severity classification.

Model	Accuracy	Precision	Recall	F1 Score
CNN	0.68	0.72	0.68	0.69
Logistic Regression	0.72	0.73	0.72	0.72
Random Forest	0.70	0.70	0.70	0.70
Gradient Boosting	0.74	0.75	0.74	0.74
SVC	0.63	0.63	0.63	0.63
XGBoost	0.72	0.72	0.72	0.72
LightGBM	0.71	0.72	0.71	0.71
MLP	0.66	0.68	0.66	0.67
Proposed Model	0.89	0.89	0.89	0.89

complexity and software tools required for model construction and evaluation. This study highlights the significance of high-performance computing systems with GPU capabilities. Intel Core i7 meets the substantial computing requirements of extensive data processing. This configuration enables concurrent processing, investigating intricate neural network topologies while reducing hardware limitations. The framework utilizes the Python programming language and advanced libraries and tools. TensorFlow is used for practical GPU-accelerated calculations on Google Colab, whilst Keras facilitates the streamlined building and training of deep learning models. This integrated framework ensures that the study provides dependable and significant insights into the severity of Parkinson's disease. Moreover, it enhances the forefront of deep learning and explainable AI, promoting advancements in both precision and clarity.

Table 10 illustrates that each element of the proposed PSS-Net pipeline significantly improved performance. The elimination of any step, including PCA or class balancing, resulted in a significant decrease in accuracy, so confirming the efficacy of our integrated system.

The proposed model surpasses all current techniques by a significant margin, indicating its potential as a reliable tool for PD severity categorization. Traditional ML models, such as Gradient Boosting and XGBoost, offer solid foundations but lack the adeptness of the proposed additions. Table 8 and 7 highlight the performance of several machine learning models for PD severity classification, as evaluated by accuracy, recall, precision, and F1 score. Traditional models such as Logistic Regression, Random Forest, and Gradient Boosting perform moderately well, with Gradient Boosting attaining the best accuracy at 74%. Models such as CNN and MLP perform marginally worse, showing difficulties managing high-dimensional and unbalanced data. The Support Vector Classifier (SVC) has the lowest performance across measures, most likely because of its sensitivity to class imbalances. Advanced boosting approaches, such as XGBoost and LightGBM, provide similar and consistent performance, demonstrating their capacity to handle complicated data structures. Notably, the suggested augmented OptiFusionNet model outperforms other models, scoring 89% across all criteria. The proposed model proves its greater capacity to

TABLE 9. Multiclass performance metrics of ML and DL models for parkinson's disease severity classification.

Model	Multiclass	Precision	Recall	F1 Score
CNN	Healthy Control (0)	0.77	0.71	0.74
	Mild (1)	0.75	0.72	0.73
	Moderate (2)	0.52	0.79	0.62
	Severe (3)	0.83	0.57	0.67
	Overall	0.72	0.68	0.69
Logistic Regression	Healthy Control (0)	0.79	0.68	0.73
	Mild (1)	0.72	0.92	0.81
	Moderate (2)	0.59	0.63	0.61
	Severe (3)	0.80	0.73	0.76
	Overall	0.73	0.72	0.72
Random Forest	Healthy Control (0)	0.69	0.76	0.72
	Mild (1)	0.73	0.76	0.75
	Moderate (2)	0.61	0.61	0.61
	Severe (3)	0.76	0.69	0.72
	Overall	0.70	0.70	0.70
Gradient Boosting	Healthy Control (0)	0.77	0.63	0.70
	Mild (1)	0.68	0.84	0.75
	Moderate (2)	0.65	0.82	0.72
	Severe (3)	0.86	0.71	0.77
	Overall	0.75	0.74	0.74
SVC	Healthy Control (0)	0.64	0.66	0.65
	Mild (1)	0.56	0.60	0.58
	Moderate (2)	0.53	0.45	0.49
	Severe (3)	0.72	0.76	0.74
	Overall	0.63	0.63	0.63
XGBoost	Healthy Control (0)	0.74	0.76	0.75
	Mild (1)	0.64	0.72	0.68
	Moderate (2)	0.64	0.66	0.65
	Severe (3)	0.80	0.73	0.76
	Overall	0.72	0.72	0.72
LightGBM	Healthy Control (0)	0.80	0.74	0.77
	Mild (1)	0.66	0.84	0.74
	Moderate (2)	0.59	0.58	0.59
	Severe (3)	0.77	0.73	0.75
	Overall	0.72	0.71	0.71
MLP	Healthy Control (0)	0.79	0.61	0.69
	Mild (1)	0.58	0.88	0.70
	Moderate (2)	0.55	0.58	0.56
	Severe (3)	0.76	0.67	0.71
	Overall	0.68	0.66	0.67
Proposed Model (PSS-Net)	Healthy Control (0)	0.90	0.79	0.84
	Mild (1)	0.86	0.88	0.87
	Moderate (2)	0.85	0.92	0.89
	Severe (3)	0.93	0.98	0.96
	Overall	0.89	0.89	0.89

TABLE 10. Ablation study of the proposed PSS-Net model.

Experiment	Accuracy(%)	Remarks
PSS-Net Model (Proposed)	0.89	Attained better accuracy with all components activated.
Without Quantile Percentile Conversion	0.83	Performance declined, indicating the significance of statistical distribution.
Without PCA	0.81	Retaining all characteristics reduced performance due to feature redundancy.
Without KMeansSMOTE (Class Balancing)	0.79	Accuracy declined; the model exhibited bias towards predominant groups.
Without TreeSelect (Feature Selection)	0.77	Using all features led to overfitting and generalization loss.
Replace OptiFusionNet with Basic CNN	0.68	The decline in accuracy was substantial, validating the preeminence of fusion design.

handle data problems and give accurate stage classification for PD severity stages.

Table 9 shows the performance metrics (Precision, Recall, and F1 Score) of several Machine Learning (ML) and Deep Learning (DL) models for categorizing Parkinson's disease (PD) severity into four categories: Healthy Control (0), Mild (1), Moderate (2), and Severe (3). It also contains an overview of each model's performance, noting its strengths and weaknesses in handling this stage classification problem. The CNN model performs well for healthy and mild instances, with high precision and recall, but suffers with moderate and severe classes, which have lower recall and F1 scores, suggesting difficulties in properly recognizing these categories. Logistic

Regression shows balanced performance, excelling in Mild instances with high recall and F1 scores while demonstrating modest success in other classes. Random Forest and Gradient Boosting provide comparable outcomes across all classes, with Gradient Boosting marginally outperforming Random Forest overall. SVC has lower metrics, demonstrating its limits for this task. The PSS-Net model surpasses all existing techniques, with a total F1 score of 89%. ML methods, such as Logistic Regression and Gradient Boosting, perform better but struggle with Moderate and severe classes. The proposed improvements to the OptiFusionNet and robust preprocessing result in a very effective solution for classifying Parkinson's disease severity. Table 10 illustrates that each element

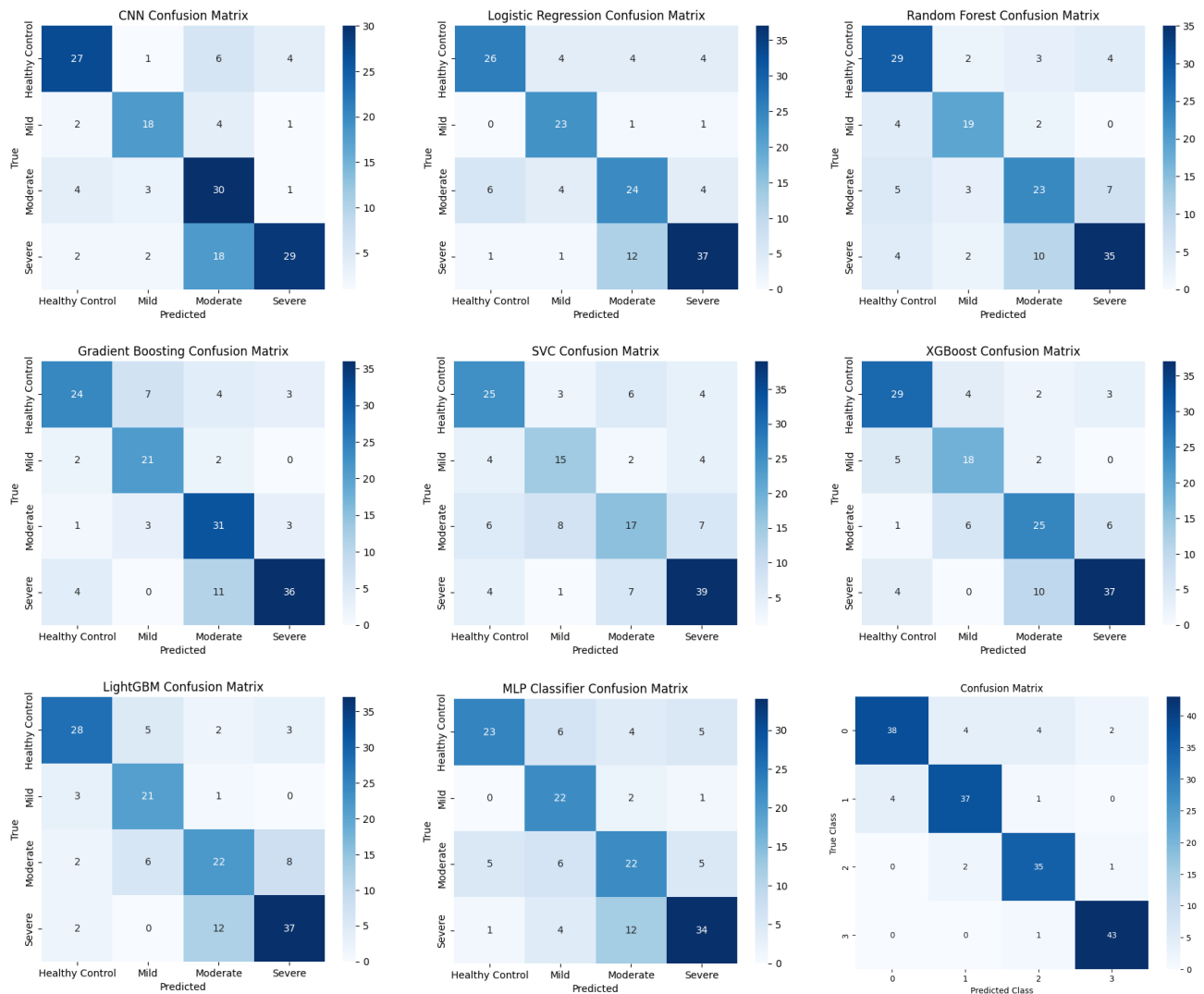


FIGURE 5. Confusion matrix analysis for parkinson's disease severity classification.

of the proposed PSS-Net pipeline significantly improved performance. The elimination of any step, including PCA or class balancing, resulted in a significant decrease in accuracy, so confirming the efficacy of our integrated system.

Figure 5 shows the confusion matrix for multiple machine learning models applied to a multi-class classification problem, which categorizes occurrences into four classes: “Healthy Control,” “Mild,” “Moderate,” and “Severe.” The models include CNN, Logistic Regression, Random Forest, Gradient Boosting, SVC, XGBoost, LightGBM, MLP Classifier, and a PSS-Net model. Advanced models like CNN, Gradient Boosting, XGBoost, and LightGBM exhibit balanced performance. The proposed PSS-Net model outperforms other techniques tested for stage classification, showcasing its superior accuracy and effectiveness. It emphasizes the model’s strong performance and promise as a dependable tool for determining Parkinson’s disease severity. PSS-Net outperforms in all four classes, with many properly

categorized cases along the diagonal. The model’s capacity to reliably categorize examples is evident, especially in Class 3, which has the most significant accuracy with the fewest misclassifications. However, there is considerable misunderstanding between Class 0 and Class 1 since each occurrence is infrequently misclassified in both directions. Despite this, the overall categorization accuracy is encouraging, with most occurrences accurately allocated to their appropriate classes. This confusion matrix provides valuable information about PSS-Net’s strengths, such as its effective categorization of Class 3. It identifies opportunities for improvement, such as minimizing misclassification between Classes 0 and 1.

Figure 6 depicts the Receiver Operating Characteristic (ROC) curve for stage classification, demonstrating the model’s ability to discriminate between various Parkinson’s disease severity classes (0, 1, 2, and 3). The ROC curve compares the True Positive Rate (TPR) to the False Positive Rate (FPR) at different classification thresholds, with the

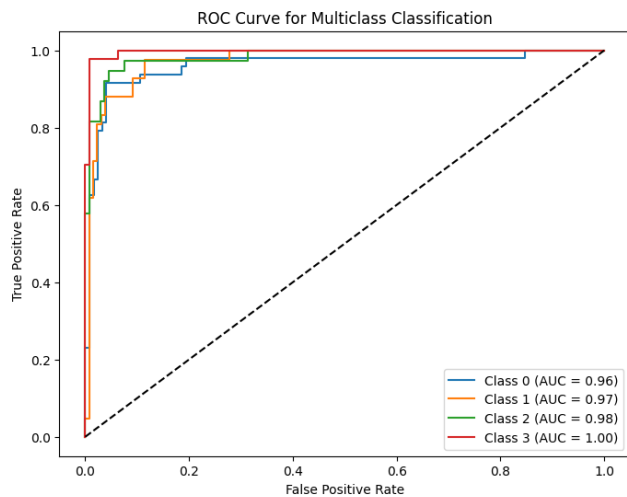


FIGURE 6. ROC for PSS-Net model.

diagonal dotted line representing the random guessing baseline ($TPR = FPR$). The model's ability to distinguish between classes improves as the ROC curves move away from the diagonal and toward the top-left corner. Figure 6 displays the AUC values for each class, which range from 0.96 to 1.0, with higher values indicating better classification performance. Class 3 has an AUC of 1.0, indicating flawless classification, and the remaining classes also perform well. Overall, the ROC curve demonstrates the PSS-Net model's better classification capabilities across all severity stages of Parkinson's disease, confirming its efficacy in this multiclassification task.

Figure 7 shows the proposed model's training and validation performance, emphasizing accuracy and loss. Both curves rise sharply in the early epochs in the training and validation accuracy plots, showing successful learning. However, although training accuracy continues to rise, achieving almost 100%, validation accuracy plateaus at epoch 50 and drops significantly toward the end, indicating probable overfitting. The training and validation loss plots suggest that training and validation losses initially decrease, indicating that model predictions are improving. However, after epoch 50, the validation loss plateaus and slightly increases, corroborating the overfitting finding. Early stopping was utilized to monitor validation loss and end training if no progress was evident after 10 consecutive epochs, ensuring that better model weights were restored. In addition, a learning rate scheduler was used to dynamically change the learning rate, lowering it by 0.2 when validation performance plateaued. These techniques served to improve the model's performance. It is observed that the proposed model showed superior performance compared to the other models. Moreover, the proposed model performs better on the validation set, highlighting the importance of early halting and dynamic learning rate modifications in improving overall performance.

Figure 8 shows a correlation matrix that depicts the SHAP interaction values of different features in an PSS-Net

model. The matrix includes the first four essential features from feature selection: `tqwt_energy_dec_18`, `f1`, `DFA`, and `PPE`. SHAP interaction values quantify the amount and direction of interactions between these features, explaining how they affect the model's predictions. In the matrix, deeper blue hues reflect negative interactions, whereas darker red colors represent positive interactions. The most significant interactions between `tqwt_energy_dec_18`, `f1`, `DFA`, and `PPE` indicate that these feature combinations considerably influence the model's output. Weak interactions are also observed between the remaining feature pairings. This correlation matrix helps to understand the relation between features and how they interact to influence the model's behavior. Analyzing these interactions provides insights into the model's decision-making process and identifies possible areas for improvement, resulting in improved model performance and interpretability. Understanding these feature interactions is critical for refining the model and making more accurate, dependable predictions in real-world applications.

B. COMPARATIVE ANALYSIS OF FEATURE SELECTION ALGORITHMS

Initially, the analysis focuses on binary classification tasks, such as discriminating between Parkinson's Disease (PD) patients and healthy controls, as shown in Table 11. The proposed model distinguishes itself by solving the more complicated stage classification problem of classifying PD severity into four stages: 0 (Healthy Control), 1 (Mild), 2 (Moderate), and 3 (Severe). Dataset consistency is maintained across all research works, including the proposed work. This ensures that the comparison focuses on methodological advances rather than data source differences. Existing research works incorporated traditional methods like normalization, standardization, SMOTE, or random oversampling to address imbalanced datasets. In contrast, the proposed model uses advanced techniques like Isolation Forest, PCA, and KMeansSMOTE, significantly improving data representation and balance for multiclass tasks. Regarding feature selection and explainability, much prior research either skips feature selection methodologies or employs simple approaches such as MFCC, PCA, or mRMR. In contrast, the PSS-Net model features tree-based feature selection and SHAP (SHapley Additive Explanations), providing greater interpretability and insights into model choices. Finally, while most existing studies use traditional classifiers such as RF and SVM, the proposed model uses an OptiFusionNet with batch normalization, which is designed explicitly for stage classification, resulting in better performance metrics across all PD severity stages. This detailed comparison demonstrates the proposed model's creativity and efficacy, notably in dealing with the issues of PD severity stage classification.

Table 11 shows the comparative analysis of the proposed PSS-Net model's improvements over existing related works in Parkinson's Disease (PD) classification methods. While all previous studies have focused solely on binary classification

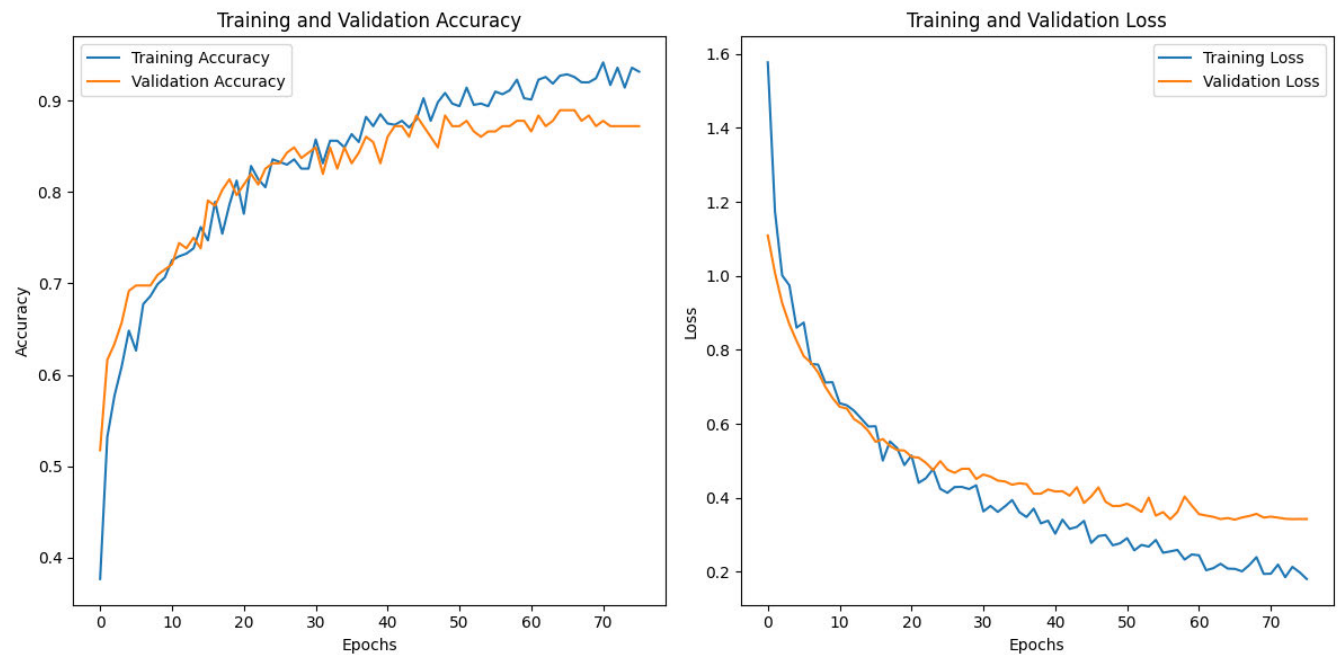


FIGURE 7. Comparative analysis of training and validation accuracy and loss for PSS-Net model.

TABLE 11. Comparison of proposed and existing classification approaches for parkinson’s disease severity prediction using UCI repository dataset.

Related Work	Task Type	Preprocessing Method	Feature Selection Method	Explainable AI (XAI)	Classification Method	Binary Accuracy (%)	Multiclass Metrics
Hawi et al. [35]	Binary Classification	Normalization and Standardization	MFCC	No	Random Forest	88.84	Not applicable
Rohit Lamba et al. [34]	Binary Classification	None	MIRFE	No	Random Forest, XGBoost	93.88	Not applicable
Mithun Shivakoti et al. [55]	Binary Classification	Random Oversampling	None	No	CatBoost	95.1	Not applicable
Nutan Singh et al. [42]	Binary Classification	SMOTE	Ensemble Classifier	No	EFS Algorithm	90.2	Not applicable
Bukhari et al. [21]	Binary Classification	SMOTE	PCA	No	AdaBoost	96	Not applicable
Sakar et al. [56]	Binary Classification	Standardization (z-score normalization)	Minimum redundancy-maximum relevance (mRMR)	No	KNN, LR, NB, RF, MLP, SVM, Ensemble Voting and Stacking	86	Not applicable
Gunduz et al. [27]	Binary Classification	Min-Max scaling after normalization	Convolutional Neural Network	No	SVM	86.9	Not applicable
Proposed PSS-Net Model	Stage classification (PD Severity Stages: 0, 1, 2, 3)	Isolation Forest, PCA, KMeans-SMOTE	Tree-based Model (Random Forest)	SHAP	OptiFusionNet with Batch Normalization	Not applicable	Healthy Control (0): Accuracy 90%, Precision 79%, Recall 84% Mild (1): Accuracy:86%, Precision:88%, Recall:87% Moderate (2): Accuracy:85%, Precision:92%, Recall:89% Severe (3): Accuracy:93%, Precision:98%, Recall:96%

(differentiating between PD patients and healthy controls), the PSS-Net model takes a novel approach by addressing the more complex task of stage classification to categorize PD severity stages (Healthy Control, Mild, Moderate, and Severe). PSS-Net uses sophisticated preprocessing methods such as Isolation Forest, PCA, and KMeansSMOTE to provide balanced data representation and effective feature extraction. Unlike previous techniques that lacked explainability or feature selection refinement, PSS-Net combines tree-based feature selection models with SHAP (SHapley Additive Explanations) to improve interpretability and model performance. Furthermore, by combining an OptiFusionNet with batch normalization, the proposed model achieves precise multiclass metrics like 90% accuracy for Healthy Control and 93% for Severe patients, demonstrating its usefulness in addressing the issues of PD severity classification. This novel technique establishes a new standard for future Parkinson’s disease monitoring.

In summary, the PSS-Net model outperforms conventional techniques in assessing Machine Learning (ML) and Deep

Learning (DL) models for classifying Parkinson’s Disease (PD) severity. With its exceptional accuracy and interpretability, the PSS-Net model establishes a new standard for classifying PD severity. Its sophisticated preprocessing, feature selection and model improvements overcome the drawbacks of conventional models, making it an effective instrument for early intervention tactics and clinical decision-making. Proficiency in explainable AI approaches and stage classification improves PD monitoring. For broader clinical usefulness, future studies might concentrate on improving PSS-Net, decreasing misclassifications in comparable severity stages, and using the model on larger, more varied datasets.

C. DISCUSSION

The advantage of the stage classification dataset is moving away from binary labels and toward multiclass ones (e.g., Mild, Moderate, Severe, and Healthy Control), which allows for a more nuanced view of disease development, improving interpretability. When PD severity fluctuates across phases in real-world circumstances, multiclass labels work better.

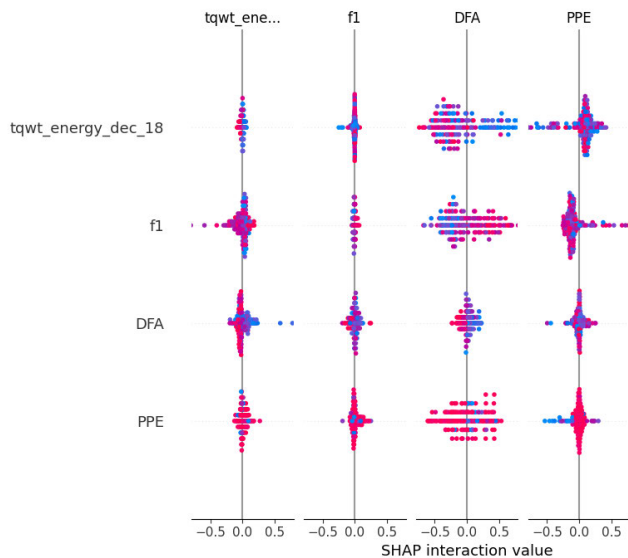


FIGURE 8. SHAP interaction matrix.

Stage classification models have a greater chance of capturing subtle patterns in the data, which might improve model training and ultimately lead to better generalization. Feature prioritization makes the model more sensitive to features that suggest severity and weights emphasize crucial aspects. Quantile-based staging levels that align with clinical standards can be applied to improve the data-driven framework's medical applicability. Selecting weights for features without clinical validation could lead to bias. Multiclassification, unlike binary classification, makes model construction and assessment more complicated. The current study primarily focuses on the stage classification of Parkinson's disease using speech features and deep learning techniques. Our objective was to explore the potential of deep learning for identifying different severity levels (e.g., Mild, Moderate, Severe) rather than developing a real-time diagnostic system. We envision that the proposed model could assist clinicians in research settings, particularly for speech-based stage classification, which may support studies related to treatment response, drug efficacy, or monitoring disease progression.

The complexity difficulties of the proposed model have been recognized and are mitigated by evaluating the impact of each component on overall performance. The pipeline comprises several critical steps: Isolation Forest for outlier identification, KMeans-SMOTE for class balance, PCA for dimensionality reduction, and Random Forest for feature selection, all of which enhance data representation and model generalization. To elucidate the trade-offs, we have incorporated a comparative analysis with traditional machine learning models such as SVM, Logistic Regression, and Random Forest, which typically present reduced computational costs but may inadequately capture intricate, nonlinear relationships inherent in the dataset, as shown in Table 8. This comparison highlights the equilibrium between the increased

methodological complexity and the resulting performance improvements, as illustrated in Table 11.

Speech features are a promising biomarker for PD staging, but have drawbacks. Although it can be beneficial for monitoring specific aspects of disease progression, particularly motor symptoms associated with speech, it is not a comprehensive or entirely accurate method for monitoring the stages of PD. In future expansions of our research, we aim to include speech data with supplementary modalities, including neuroimaging and clinical assessments, such as the Unified Parkinson's Disease Rating Scale (UPDRS) and Hoehn and Yahr (H&Y) staging. This multimodal integration, along with longitudinal monitoring, is anticipated to improve the reliability and accuracy of Parkinson's disease stage categorization. We want to investigate temporal modeling approaches, including recurrent neural networks (RNNs), memory-augmented transformers, and continuous learning frameworks, to provide dynamic and adaptive modeling across time. These developments will facilitate the creation of lifespan-aware classification algorithms, essential for practical use in chronic progressive illnesses such as Parkinson's disease. Secondly, the use of quantile-based label creation may result in class overlap, which might adversely impact classification accuracy. Finally, despite implementing regularization techniques such as dropout and early stopping, the model still exhibits signs of overfitting, particularly when it adopts overly ambitious and deeper architectures. Therefore, to ensure more reliable tracking of PD progress, the existing technique may be confirmed and augmented with supplementary data, notwithstanding its promise as a helpful tool for disease monitoring.

V. CONCLUSION AND FUTURE WORK

This work introduces a data-driven deep learning framework for stage classification of Parkinson's Disease (PD) using speech data, filling a significant gap in previous research works mainly concentrating on binary classification. The PSS-Net model effectively identified the most essential speech features for PD staging using sophisticated pre-processing approaches such as Isolation Forest for outlier identification, KMeansSMOTE for class balancing, and Random Forest-based feature selection. The Optimized and Fused Neural Network (OptiFusionNet) attained 89% accuracy, indicating its potential for stage-specific classification. Furthermore, the SHAP technique offers insights into feature contributions and increases clinician confidence. To begin with, it provides stage-specific insights for treatments, closing a crucial knowledge vacuum in the field by expanding beyond binary categorization to staging. The second reason is that physicians can comprehend how certain speech variables impact the categorization thanks to SHAP (SHapley Additive exPlanations), which makes the model interpretable and increases confidence and practicality. Lastly, the scalable, non-invasive, and transparent model can revolutionize PD by offering a tailored treatment plan, early identification, and cost-effectiveness.

This study establishes the groundwork for future research to integrate multimodal data and explore new architectures for more significant clinical usage. It also demonstrates the effectiveness of speech data in PD staging. Future research might build on this work using more critical and varied datasets to increase model generalizability and robustness. The integration of multimodalities, including speech and other biomarkers such as gait or neuroimaging data, could enhance classification accuracy and reliability. The PSS-Net can be enhanced further to address other neurodegenerative disorders, expanding its utility beyond Parkinson's Disease. Furthermore, investigating new deep learning architectures such as Transformer models or Graph Neural Networks may enhance feature extraction and classification performance. Future work may involve collaborating with clinical experts to obtain specialist-annotated labels, facilitating the assessment of inter-rater variability. Finally, the current model is trained on a curated speech dataset from the UCI repository, but for clinical deployment, more diverse and representative data is needed. Future extensions will incorporate non-invasive modalities like handwriting analysis or wearable sensor data. Interpretability of results is crucial for clinical trust and adoption. The study used SHAP to identify key speech features, but future work will use SHAP-guided feature pruning, validate model findings against clinical knowledge, and include expert feedback. Deployment strategy involves implementing model compression techniques to reduce computational load and enable deployment on mobile or embedded systems. Continuous learning and temporal modeling will ensure long-term clinical usability. Building real-time, deployable solutions for clinical usage would bridge the gap between research and practice, allowing for earlier and more accurate PD prediction and staging.

DECLARATION OF COMPETING INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY

The data used in this study are from publicly available datasets. The datasets used are as follows: <https://archive.ics.uci.edu/dataset/470/parkinson+s+disease+classification>.

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K. SHYAMALA is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Vellore Institute of Technology (VIT), Tamil Nadu, India. Her research interests include machine learning and deep learning.



T. M. NAVAMANI received the M.E. degree in CSE and the Ph.D. degree from Anna University, Chennai. She is currently a Senior Associate Professor with the School of Computer Science and Engineering, Vellore Institute of Technology (VIT), Vellore, India. She has 27 years of teaching experience. She has published more than 45 papers in international and national journals, conferences, and book chapters. Her research interests include data sciences, human-computer interaction, wireless ad hoc networks, network security, and mobile computing. She is a Life Member of Indian Society for Technical Education (ISTE).

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